Definition of the equilibrium

A competitive equilibrium is a price system $\{P_1, P_2, Q\}$ and allocations $\{C_1, C_2, X, K\}$ such that

i) The consummer maximizes the intertemporal utility given prices and subject to the intertemporal budget constraint.

ii) The firm maximizes profit given prices and the technology.

iii) Supply=Demand for each goods

$$C_1 + X = Y_1 \quad \text{(good market in period 1)}$$

$$C_2 = Y_2 + F(K) \quad \text{(good market in period 2)}$$

$$X = K \quad \text{(capital market)}$$